EE133 Solid-State Electronics

Homework 6

Kaushik Vada

February 23, 2026

Given: Si pn junction with $N_A = 10^{15}$ cm^{-3}, $N_D = 10^{17}$ cm^{-3}, $T = 300$ K, $n_i = 10^{10}$ cm^{-3}.
Constants: $\varepsilon_{Si} = 11.7 \times 8.854 \times 10^{-12}$ F/m $= 1.036 \times 10^{-12}$ F/cm, $q = 1.6 \times 10^{-19}$ C, $k_B T/q = 0.02585$ V.

Problem 1 — Zero Bias

(a) Built-in Voltage V_{bi}

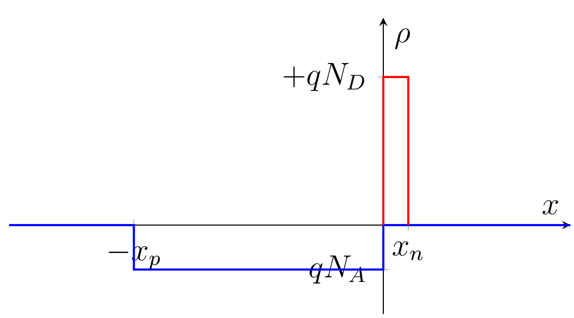
$$V_{bi} = \frac{k_B T}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right) = (0.02585)\ln\left(\frac{10^{15} \times 10^{17}}{(10^{10})^2}\right) = (0.02585)\ln(10^{12}) \boxed{= 0.714 \text{ V}} \tag{1}$$

(b) Charge Density (depletion approximation)

Under the depletion approximation the charge density is a two-step rectangle:

$$\rho(x) = \begin{cases} -qN_A = -1.6 \times 10^{-4} \text{ C/cm}^3 & -x_p \le x \le 0 \\ +qN_D = +1.6 \times 10^{-2} \text{ C/cm}^3 & 0 \le x \le x_n \\ 0 & \text{otherwise} \end{cases}$$

Because $N_D \gg N_A$ the positive rectangle is 100× taller but correspondingly ∼100× narrower (charge neutrality requires $N_A x_p = N_D x_n$).



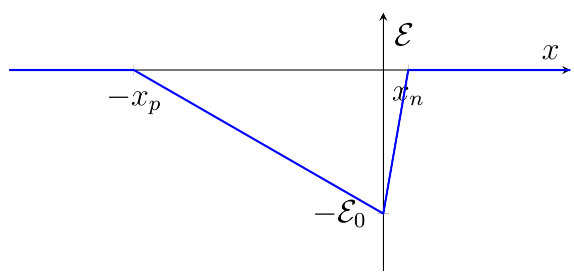
(c) Two Expressions for Maximum Electric Field $\mathcal{E}_0$

Integrating Gauss's law ($d\mathcal{E}/dx = \rho/\varepsilon_{Si}$) from each edge of the depletion region to $x = 0$:

$$\boxed{\mathcal{E}_0 = \frac{qN_A x_p}{\varepsilon_{Si}}} \qquad \text{and} \qquad \boxed{\mathcal{E}_0 = \frac{qN_D x_n}{\varepsilon_{Si}}} \tag{2}$$

(d) Electric Field Profile

The field is piecewise linear, peaking in magnitude at $x = 0$ and returning to zero at $x = -x_p$ and $x = x_n$. It points in the $-x$ direction (from n toward p).



(e) Expression for V_{bi} from the Electric Field

V_{bi} equals the area of the triangular electric-field profile:

$$\boxed{V_{bi} = \frac{1}{2}\,\mathcal{E}_0\,(x_n + x_p) = \frac{1}{2}\,\mathcal{E}_0\,W} \tag{3}$$

(f) Total Depletion Width W

Starting from charge neutrality ($N_A x_p = N_D x_n$) and the potential condition $V_{bi} = \frac{1}{2}\mathcal{E}_0 W$:

$$W = \sqrt{\frac{2\varepsilon_{Si}}{q}\,V_{bi}\,\frac{N_A + N_D}{N_A N_D}} = \sqrt{\frac{2(1.036 \times 10^{-12})}{1.6 \times 10^{-19}}\,(0.714)\,\frac{1.01 \times 10^{17}}{10^{15} \times 10^{17}}} \tag{4}$$

$$W = \sqrt{9.32 \times 10^{-9}}\,\boxed{= 9.65 \times 10^{-5}\text{ cm} = 0.965\ \mu\text{m}}$$

(g) Depletion Width on the n-side, x_n

$$x_n = W\,\frac{N_A}{N_A + N_D} = (9.65 \times 10^{-5})\,\frac{10^{15}}{1.01 \times 10^{17}}\,\boxed{= 9.55 \times 10^{-7}\text{ cm} \approx 9.55\text{ nm}} \tag{5}$$

This is very small because $N_D \gg N_A$ — almost all the depletion is on the lightly doped p-side.

(h) Depletion Width on the p-side, x_p

$$x_p = W \frac{N_D}{N_A + N_D} = W - x_n \approx 9.65 \times 10^{-5} \text{ cm} \boxed{= 0.956\ \mu\text{m}} \tag{6}$$

(i) Voltage at $x = 0$

Taking $V(-x_p) = 0$, the potential at $x = 0$ equals the area under $|\mathcal{E}|$ from $-x_p$ to 0 (the p-side triangle):

$$V(0) = \frac{qN_A x_p^2}{2\varepsilon_{Si}} = \frac{(1.6 \times 10^{-19})(10^{15})(9.56 \times 10^{-5})^2}{2(1.036 \times 10^{-12})} \boxed{= 0.707 \text{ V}} \tag{7}$$

As a sanity check, the remaining $V_{bi} - V(0) = 0.007$ V is the voltage drop across the narrow n-side. ✓

(j) Junction Capacitance C_j (per unit area)

$$C_j = \frac{\varepsilon_{Si}}{W} = \frac{1.036 \times 10^{-12}}{9.65 \times 10^{-5}} \boxed{= 1.073 \times 10^{-8} \text{ F/cm}^2 = 10.73 \text{ nF/cm}^2} \tag{8}$$

Problem 2 — Reverse Bias $V_A = -1$ V

A negative (reverse) bias of $V_A = -1$ V increases the effective barrier voltage:

$$V_{\text{total}} = V_{bi} - V_A = 0.714 + 1.0 = 1.714 \text{ V}$$

All formulas are identical to Problem 1 with $V_{bi} \to V_{\text{total}}$. The charge-density *shape* is unchanged (same ρ magnitudes), but the depletion region is wider.

(b) Charge Density

Same functional form as Problem 1b:

$$\rho(x) = \begin{cases} -qN_A = -1.6 \times 10^{-4} \text{ C/cm}^3 & -x_p \le x \le 0 \\ +qN_D = +1.6 \times 10^{-2} \text{ C/cm}^3 & 0 \le x \le x_n \end{cases}$$

The widths x_n, x_p are larger (see below); sketch is qualitatively identical but the x-axis extends further.

(c) Electric Field Expressions (same form)

$$\mathcal{E}_0 = \frac{qN_A x_p}{\varepsilon_{Si}} \qquad \text{and} \qquad \mathcal{E}_0 = \frac{qN_D x_n}{\varepsilon_{Si}} \tag{9}$$

(d) Electric Field Profile

Same triangular shape, but larger $|\mathcal{E}_0|$ and wider depletion region.

(e) Total Voltage from Electric Field

$$V_{\text{total}} = \frac{1}{2}\,\mathcal{E}_0\,(x_n + x_p) = \frac{1}{2}\,\mathcal{E}_0\,W = 1.714 \text{ V} \tag{10}$$

(f) Total Depletion Width W

$$W = \sqrt{\frac{2\varepsilon_{Si}}{q}\,V_{\text{total}}\,\frac{N_A + N_D}{N_A N_D}} = \sqrt{\frac{2(1.036\times10^{-12})}{1.6\times10^{-19}}\,(1.714)\,\frac{1.01\times10^{17}}{10^{32}}}\boxed{= 1.497\times10^{-4} \text{ cm} = 1.497\ \mu} \tag{11}$$

Note: $W_{\text{rev}}/W_0 = \sqrt{1.714/0.714} = 1.550$, so the depletion width grows by 55% under -1 V reverse bias. ✓

(g) Depletion Width on n-side, x_n

$$x_n = W\,\frac{N_A}{N_A + N_D} = (1.497\times10^{-4})\,\frac{10^{15}}{1.01\times10^{17}}\boxed{= 1.482\times10^{-6} \text{ cm} \approx 14.82 \text{ nm}} \tag{12}$$

(h) Depletion Width on p-side, x_p

$$x_p = W - x_n \approx 1.497\times10^{-4} \text{ cm}\boxed{= 1.482\ \mu\text{m}} \tag{13}$$

(i) Voltage at $x = 0$

$$V(0) = \frac{qN_A x_p^2}{2\varepsilon_{Si}} = \frac{(1.6\times10^{-19})(10^{15})(1.482\times10^{-4})^2}{2(1.036\times10^{-12})}\boxed{= 1.697 \text{ V}} \tag{14}$$

The remaining $V_{\text{total}} - V(0) = 0.017$ V is the small drop across the n-side. ✓

(j) Junction Capacitance C_j (per unit area)

$$C_j = \frac{\varepsilon_{Si}}{W} = \frac{1.036\times10^{-12}}{1.497\times10^{-4}}\boxed{= 6.92\times10^{-9} \text{ F/cm}^2 = 6.92 \text{ nF/cm}^2} \tag{15}$$

Reverse bias reduces the capacitance by a factor of $C_{j,0}/C_{j,\text{rev}} = 10.73/6.92 \approx 1.55$, consistent with W growing by the same factor. ✓

Summary Table

Quantity	Problem 1 ($V_A = 0$)	Problem 2 ($V_A = -1$ V)
V_{total}	0.714 V	1.714 V
W	0.965 μm	1.497 μm
x_n	9.55 nm	14.82 nm
x_p	0.956 μm	1.482 μm
$\mathcal{E}_0$	1.478×10^4 V/cm	2.290×10^4 V/cm
$V(x = 0)$	0.707 V	1.697 V
C_j	10.73 nF/cm^2	6.92 nF/cm^2